

FLOODGAUGE · REQUIREMENTS

FloodGauge requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/03-requirements.md @ 13e3eb9		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

FloodGauge requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish and the co-design partner, and will be checked by calculation at TRL 3. The status column gives the first-order result from FLG-PRC-001; "not met" is stated plainly.

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Measure water depth on the street at the gauge point	0 to 600 mm above the road surface, head mounted 2.5 to 3.5 m above the road	Range calculation against sensor data sheet	Met on paper (radar range up to 20 m)
R2	Street depth accuracy	±10 mm over -20 to 50 °C air, still water	Error budget; later bench test over a tank	Met on paper with radar (data sheet accuracy unverified); not met by an uncompensated ultrasonic head (about ±100 mm)
R3	Measure water level in the catch basin	From 50 mm above the basin floor to the underside of the grate	Geometry of stilling tube and head	Met by design
R4	Drain level accuracy	±20 mm	Error budget	Met on paper with temperature compensation (about ±15 mm, estimate)
R5	Sampling rate	Every 60 s normally; every 10 s when level is above the first alert band or rising faster than 20 mm/min	Firmware sketch review	Met by design
R6	Alert latency	120 s or less from threshold crossing to message at the alert service, 95 % of events	Latency budget	Met on paper with a private gateway: about 70 s typical, about 110 s worst case (estimate); thin margin
R7	Radio airtime within network rules	Within EU868 1 % duty cycle always; within The Things Network fair use (30 s uplink airtime per day) in normal mode	Airtime calculation	Normal mode met (about 24 s/day); not met on The Things Network during events longer than about 24 min at 1 min reporting
R8	Energy autonomy	5 days without sun at event-mode sampling	Power budget against FieldNode	Met on paper (sensor load about 7 mW worst case against about 115 mW FieldNode allowance)
R9	Survive submersion of the drain head	IP68, 2 m, 72 h, stormwater with silt	Enclosure design review	Not met by the stock ultrasonic module (IP67); potted housing proposed
R10	Reject false readings	No alert caused by a vehicle, person or debris under the street head; target 1 false alert per gauge per year or fewer	Plausibility logic review; later field log	Unverified
R11	Privacy	Levels, status and battery only; no camera or microphone; no images or audio leave the device	Design review	Met by design
R12	Installation	Two-person crew, 90 min or less, from the surface; no road drilling; no entry into the catch basin	Installation sequence review	At risk: the conduit from basin to pole needs a core through the basin wall and a trench or bore under the sidewalk

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R13	Level datum	Head height surveyed to the road surface at the gauge point within ± 5 mm; checked yearly against the depth marker	Survey method	Met by design (method to be written at TRL 3)
R14	Operating environment	-20 to 50 °C air; UV, rain and road spray; FieldNode charges only from 0 to 45 °C	Material and cell review	Met on paper, inherited from FieldNode
R15	Mounting	Fits 40 to 60 mm poles with band clamps; total added mass on the pole 5 kg or less	Mass estimate	Met on paper (about 3.1 kg, estimate)
R16	Cost	Parts for one complete gauge \$150 or less at quantity 1	Priced BOM	Not met: about \$271 including the \$126 FieldNode core; FloodGauge-specific parts about \$145
R17	Open data	Levels published in an open, documented format (JSON or CSV) through the gateway	Design review	Met by design

■ Assumptions

- The road surface at the gauge point is the datum for street depth. Depth equals the surveyed head height minus the measured range.
- "Event" means water above the first alert band or rising faster than 20 mm/min.
- FieldNode figures (energy allowance, airtime per uplink, charge temperature window) are taken from the FieldNode README and design precis and are themselves estimates.
- Alert depths of 150 mm and 300 mm follow NWS guidance on water that can knock over an adult and carry away a car ([NWS](#)). They are proposed defaults, awaiting Amish and the partner city.

■ Requirements not met or at risk

- **R16 cost:** about \$271 against \$150. See REVIEW.md for the proposed options.
- **R9 submersion:** the stock drain sensor is IP67; the head needs potting or a sealed housing rated for longer immersion.
- **R7 airtime:** long events exceed The Things Network fair use limit; a private or city gateway is needed for event reporting.
- **R12 installation:** routing the drain head cable to the pole is civil work, not a 90 min job.
- **R2** would not be met if an ultrasonic street head were chosen instead of radar.